Blind (zero-pilot) CFO receiver

1024-symbol QPSK packet, polar code N = 2048 with CRC-16, CFO |f| ≤ 0.03 cycles / symbol

Received samples $r_x(m)$, 8 samples / symbol

Matched filter (RRC, β = 0.25) and downsampling → r_n, n = 0 ... 1023

STAGE 1: FAST PATH

4th-power CFO estimate

$y_n = -r_n^4$ (removes QPSK: a pure tone at $4f$)

$Y[k]$ = FFT of y_n (4096 points), peak in $|\nu| \le 4 \times 0.03$

parabolic interpolation → $\hat{\nu}$, $f_0 = \hat{\nu}/4$

Pre-MF correction

$r_x(m)\, e^{-j2\pi f_0 m/8}$ → matched filter again → r'_n

Carrier phase and the π ambiguity

$\theta_0 = \frac{1}{4}\arg\sum_n y'_n\, e^{-j2\pi 4 f_0 n}$ (known modulo π/2)

π/2 branch: the smaller list-1 path metric of θ_0 and $\theta_0 + \pi/2$

π branch: PiCode (the complement codeword is also accepted)

CA-SCL-8 decode

$\mathrm{LLR}_n = \sqrt{2}\, y_n/\sigma^2$ on $r'_n\, e^{-j(2\pi f_0 n + \theta_0)}$

CRC-16 (PiCode) passes and $\rho > \rho_{\mathrm{th}}(\mathrm{SNR})$? — *yes*

Packet accepted
deliver the data bits and the CFO (decision-directed refinement over the 1024 symbols)

no

STAGE 2: SEARCH

Grid of hypotheses (cells)

$f_c = -0.03 + c\,k\,\mathrm{STEP}$, $\mathrm{STEP} = 7.5 \times 10^{-5}$ (k = 4 at R = 1/10 and 0.15)

phases $\theta_c + q\,\pi/4$ (θ_c from the 4th-power FFT at $4f_c$)

Partial decodes as a CFO detector

list-1 Fast-SSC path metric M_c up to a checkpoint (½ of the code)

the frozen bits act as implicit pilots: M_c is small only near the true (f, θ)

R = 1/10, 0.15: coarse start + batches of 32 cells; R = 2/5, 1/2: checkpoint NN

Robust z-test

$z_c = \dfrac{\mathrm{median}(M) - M_c}{1.4826\ \mathrm{MAD}(M)}$

drop the cells with $z_c \le \tau$ (keep the best 16); their neighbours join

early win: $z_c > 6$? — *yes*

no

Later checkpoints (¾, 1)

repeat the path metric and the z-test

candidates: $z > 4$ (at most 4), then the best 32 by path metric

Re-encoding check

$\rho = \dfrac{|\sum_n r'_n\, \tilde{s}_n^*\, e^{-j2\pi\hat{f}n}|}{\sqrt{N\sum_n |r'_n|^2}}$

$\tilde{s}_n$: the re-encoded decision

right packet:

$\rho \approx \sqrt{\mathrm{SNR}/(1+\mathrm{SNR})}$

unrelated codeword:

$P(\rho > t) \approx e^{-Nt^2}$

$\rho_{\mathrm{th}} = \max\left(0.12,\ c(\mathrm{SNR})\sqrt{s/(1+s)}\right)$

c: the per-rate table RHO_N

STAGE 3: DECODE

Refinement of each candidate

CFO offsets $\pm k/2$ STEP, phase offsets $\{-\pi/12,\ 0,\ +\pi/12\}$

scored with the full path metric → the best T = 2 hypotheses

List decoding

R = 1/10, 0.15: CA-SCL-8 per hypothesis

R = 2/5, 1/2: one joint list of 16 paths over 4 cells × 2 hypotheses

Code-aided retry (when the CRC fails)

SC hard decisions $\hat{s}_n$: $z_n = r'_n\, \hat{s}_n^*$, 2 Newton steps on $|\sum_n z_n e^{-j2\pi f n}|^2$

$\theta = \arg\sum_n z_n e^{-j2\pi f n}$, then one more CA-SCL-8 decode

CRC-16 (PiCode) passes and $\rho > \rho_{\mathrm{th}}(\mathrm{SNR})$? — *yes* — *retry*

no

Next candidate, then rescue

next 256 cells on both π/2 branches; half-STEP grid × π/16 phases on the best 4

search exhausted

Nothing accepted → block error